

### Exercise C1

Use the simplex algorithm to find the optimal solution to the following LP:

$$\begin{array}{ll}\text{Min } z = & -4x_1 + x_2 \\ \text{S.t.} & 3x_1 + x_2 \leq 6 \\ & -x_1 + 2x_2 \leq 0 \\ & x_1, x_2 \geq 0\end{array}$$

### Exercise C2

Use the simplex algorithm to find two optimal solutions to the following LP:

$$\begin{array}{ll}\text{Max } z = & 5x_1 + 3x_2 + x_3 \\ \text{S.t.} & x_1 + x_2 + 3x_3 \leq 6 \\ & 5x_1 + 3x_2 + 6x_3 \leq 15 \\ & x_1, x_2, x_3 \geq 0\end{array}$$

### Exercise C3

Consider the following LP:

$$\begin{array}{ll}\text{Max } z = & 10x_1 + x_2 \\ \text{S.t.} & x_1 \leq 1 \\ & 20x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0\end{array}$$

- (a) Find all the basic feasible solutions for this LP.
- (b) Show that when the simplex is used to solve this LP, every basic feasible solution must be examined before the optimal solution is found.

By generalizing this example, Klee and Minty (1972) constructed (for  $n = 2, 3, \dots$ ) an LP with  $n$  decision variables and  $n$  constraints for which the simplex algorithm examines  $2^n - 1$  basic feasible solutions before the optimal solution is found. Thus, there exists an LP with 10 variables and 10 constraints for which the simplex requires  $2^{10} - 1 = 1,023$  pivots to find the optimal solution. Fortunately, such “pathological” LPs rarely occur in practical applications.

### Exercise C4

Two different products, P1 and P2, can be manufactured by one or both of two different machines, M1 and M2. The unit processing time of either product on either machine is the same. The daily capacity of machine M1 is 200 units (of either P1 or P2, or a mix of both), and the daily capacity of machine M2 is 250 units. The shop supervisor wants to balance the production schedule of the two machines such that the total number of units produced on one machine is within 5 units of the number produced on the other. The profit per unit of P1 is \$10 and that of P2 is \$15. Set up the problem as an LP in equation form.

**Exercise P1 (Optional)**

Use the simplex algorithm to find two optimal solutions to the following LP. How many optimal solutions does this LP have? Find a third optimal solution.

$$\begin{aligned}
 \text{Max } z &= 4x_1 + x_2 \\
 \text{S.t. } 2x_1 + 3x_2 &\leq 4 \\
 x_1 + x_2 &\leq 1 \\
 4x_1 + x_2 &\leq 2 \\
 x_1, x_2 &\geq 0
 \end{aligned}$$

**Exercise P2 (Optional)**

Show how the following objective function can be presented in equation form:

$$\begin{aligned}
 \text{Minimize } z &= \max \{|x_1 - x_2 + 3x_3|, |-x_1 + 3x_2 - x_3|\} \\
 x_1, x_2, x_3 &\geq 0
 \end{aligned}$$

**Exercise P3 (Optional)**

Show that the  $m$  equations

$$\sum_{j=1}^n a_{ij}x_j = b_i, \quad i = 1, 2, \dots, m$$

are equivalent to the following  $m + 1$  inequalities:

$$\sum_{j=1}^n a_{ij}x_j = b_i, \quad i = 1, 2, \dots, m$$

$$\sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} \right) x_j \geq \sum_{i=1}^m b_i$$

**Exercise P4 (Optional)**

During the 1972 football season, the games shown in Table below were played by the Miami Dolphins, the Buffalo Bills, and the New York Jets. Suppose that on the basis of these games, we want to rate these three teams. Let  $M$  = Miami rating,  $J$  = Jets rating, and  $B$  = Bills rating. Given values of  $M$ ,  $J$ , and  $B$ , you would predict that when, for example, the Bills play Miami, Miami is expected to win by  $M - B$  points. Thus, for the first Miami-Bills game, your prediction would have been in error by  $|M - B - 1|$  points. Show how linear programming can be used to determine ratings for each team that minimize the sum of the prediction errors for all games.

| Miami | Bills | Jets |
|-------|-------|------|
| 27    | -     | 17   |
| 28    | -     | 24   |
| 24    | 23    | -    |
| 30    | 16    | -    |
| -     | 24    | 41   |
| -     | 3     | 41   |